

Sensing-Fault-Aware Reinforcement Learning for Droplet Routing on Micro-Electrode-Dot-Array Biochips

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Abstract

Prior reinforcement-learning routers for MEDA biochips are fault-aware for the fluidic layer but assume a fault-free sensing layer, trusting capacitive health maps that can themselves suffer noise, spoofing, dropout, and delay. This interim report documents the learnability milestone of a sensing-fault-aware alternative: a Double-DQN router that receives a per-cell residual-based confidence signal alongside reported health and learns how much to trust each reading, evaluated in a fully digital simulator with architecturally decoupled fluidic and sensing fault channels. We report a failed first full-budget run honestly — sparse rewards and vanilla DQN dynamics produced baseline loss divergence (peaking at 12441) against aware-agent value collapse — and the minimal, identically applied fixes that restored learning: potential-based shaping on reported positions, Double-DQN target, and Huber loss. Gated clean-sensing pilots over 1200 episodes give GREEN (confidence-aware probe plateau 0.42→0.56; baseline churning near 0.40 with converged losses), supported by bounded TD losses, healthy Q-spread, and verified greedy evaluation. No fault-robustness gap is claimed: faulty-phase training and the four-experiment battery (parity, fault sweep, ablation, co-occurrence) are the wired-up next milestone.

1 Introduction

Digital microfluidic biochips (DMFBs) manipulate discrete droplets across an array of actuated electrodes to execute biochemical assays. Microelectrode-dot-array (MEDA) architectures extend this paradigm with fine-grained, freely addressable microcells, enabling eight-directional droplet transport rather than the Manhattan routing of conventional electrode arrays [1]. Routing droplets efficiently around faulty cells is therefore a sequential decision-making problem, and recent work has applied reinforcement learning (RL) to it with promising results.

The strongest such line, exemplified by prior deep RL routing [1], trains RL routers that are *fault-aware for the fluidic layer*: the agent observes a real-time health map of the array and learns to route around dead or degraded electrodes. This report starts from a gap in that line of work: the sensing layer itself is assumed fault-free. The health map the agent trusts is produced by capacitive sensing circuitry that can itself suffer noise, spoofed readings, signal dropout, and latency — and a fluidic fault masked by a simultaneous sensing fault is precisely the case the trust-the-map agent handles worst. To our knowledge, no prior RL routing work models sensing faults, let alone the co-occurrence of fluidic and sensing faults on the same cell.

Our contribution is a sensing-fault-aware router: alongside the (possibly corrupted) health reading, the agent receives a per-cell *confidence signal* derived from the residual between the commanded (expected) droplet position and the reported position, smoothed over a rolling window. The agent *learns* how much to trust each reading; there is no hard-coded cutoff. The baseline conditions differ only in the absence of this signal, so any performance gap is attributable to learned distrust rather than capacity. All experiments are fully digital: array, droplets, faults, and sensors are simulated, with fluidic and sensing fault channels architecturally decoupled.

This is an *interim* report. Its scope is the learnability milestone that precedes the headline comparison: we show (i) that the first full-budget training run failed and why, via logged evidence; (ii) the minimal, claim-preserving fixes; and (iii) gated pilot evidence that both agents learn under the fixed setup over 1200 clean-sensing episodes. The fault-robustness comparison (clean parity, sensing-fault degradation curves, confidence ablation, co-occurrence) awaits the full retrain and is explicitly out of scope here. Section 3 describes the methodology, Section 4 the implementation and the 1200-episode results, and Section 5 the path to the full evaluation.

2 Literature

2.1 MEDA architecture

Micro-electrode-dot-array (MEDA) biochips, first introduced in [2], place a CMOS control and sensing layer beneath every microelectrode, giving per-cell actuation and capacitive readout at fine pitch. Figure 1 shows the scale of this integration: a packaged chip (top left), the excitation/readout circuits tiled across the array (a), the sensing-pixel schematic with its transimpedance-amplifier readout (b), and the electrode profile and cross-section (c). This density enables free-direction routing and fine-grained droplet control — and a fault space strictly broader than that of conventional EWOD chips, surveyed below. Faults divide throughout into *catastrophic (hard)* faults, causing total malfunction, and *parametric (soft)* faults, degrading performance until a tolerance threshold is crossed.

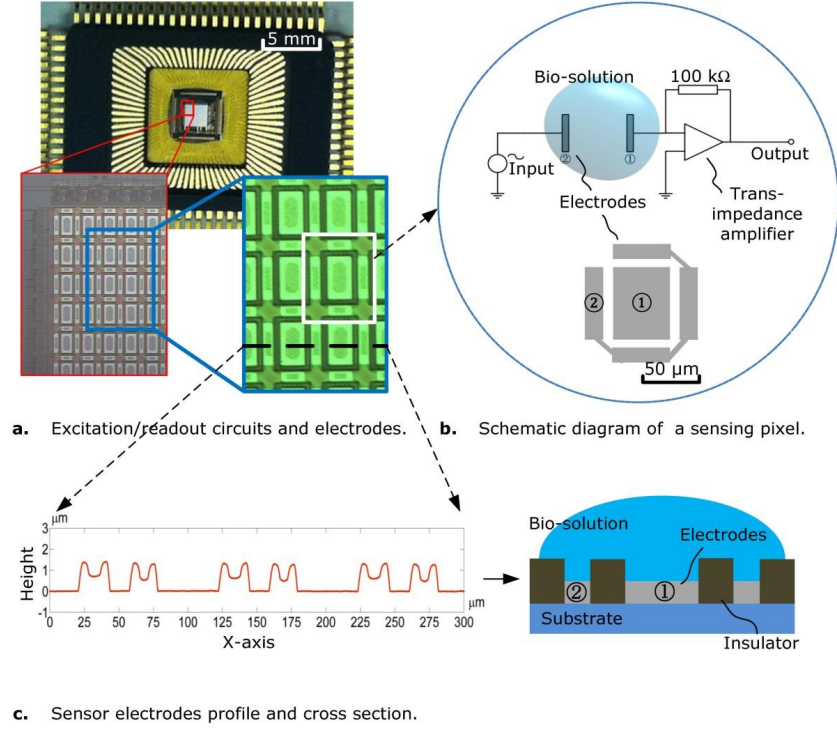


Figure 1: MEDA sensing integration: packaged chip with excitation/readout circuits (a), sensing-pixel schematic with transimpedance-amplifier readout (b), and electrode profile/cross-section (c). Every micro-cell carries its own sense circuitry — the same circuitry whose failures motivate this work.

2.2 Error modes and faults in MEDA biochips

Subsections below are ordered by descending volume of published research on each layer, with sensing-circuit faults last: they are the least modeled as a first-class failure mode, and this work’s target. Original letter IDs are kept as stable tags.

2.2.1 Catastrophic microelectrode-level faults

Dielectric breakdown (high voltage shorts droplet to electrode, triggering electrolysis and halting transport), shorts between adjacent electrodes (fused cells can no longer be actuated independently), and open/broken interconnects leaving an electrode unactivatable (stuck-at). *Existing methods*: a deep RL router is trained in [1] that avoids unhealthy electrodes in real time at commercial array sizes (advantage: online adaptation, longer chip usability); multiple simultaneous array faults are detected in [3] via pebble-traversal test droplets and the array is reconfigured to perform like new (advantage: strong multi-fault coverage); modules are placed in [4] to minimize breakdown risk at synthesis time (advantage: prevention rather than reaction). *Drawbacks*: the deep RL router in [1] needs offline training plus an accurate simulator, is single-agent, and assumes its health

sensing is fault-free; the pebble-traversal scheme in [3] is offline only (whole chip idle, traversal time scales with size, no mid-assay repair); synthesis-time placement in [4] cannot react to field faults.

2.2.2 Parametric and degradation faults

Charge trapping — cumulative electrode wear from repeated actuation that weakens actuation force until droplets stick, and can progress into dielectric breakdown — plus fabrication-tolerance deviations and defect-induced droplet volume loss. *Existing methods:* explicit robustness to charge-trapping degradation is shown in [1] (advantage: handles the most common wear mechanism); degradation-aware routing is extended in [5] to concurrent multi-agent transport without an external camera (advantage: parallelism). *Drawbacks:* same simulator/single-agent and sensing-assumption limits as above; the multi-agent extension in [5] pays coordination overhead and training complexity that grows with droplet count.

2.2.3 Daisy-chain and control-path faults

All microelectrodes and control registers share one long daisy chain, so a single failed cell breaks the chain and discards the chip. *Existing methods:* functional-pattern repair is proposed in [6] and operates mid-assay with no influence on the executing bioassay and no added degradation or time (advantage: zero-downtime repair); enhanced built-in self-diagnosis/repair is presented in [7] and improves resolution and coverage over earlier BISD/BISR schemes (advantage: finer, broader chain repair). *Drawbacks:* both are scoped to the chain (blind to fluidic and sensing faults) and bounded by per-cell redundancy, with hardware overhead scaling in chip size.

2.2.4 System and bioassay-level consequences

Stuck or misrouted droplets, routing-collision cross-contamination, incorrect dilution and mixing ratios, and cascading assay-timing violations. *Existing methods:* tests for parallel fluidic modules are scheduled in [8] (advantage: module-level coverage at lower test cost); collision-avoidance routing [4] and pebble-based reconfiguration [3] operate at this level (advantage: assay-visible reliability); contamination-aware cooperative MARL routing is pursued in [9] (advantage: fewer cross-contamination events). *Drawbacks:* symmetric-module testing in [8] depends on module symmetry and irregular layouts defeat it; synthesis-time methods cannot respond mid-assay; offline reconfiguration needs an idle chip; MARL coordination adds overhead.

2.2.5 MEDA-specific advanced-operation faults

Contact-line pinning during diagonal movement and defects from variable fine-grained droplet sizing and channel-based routing — classes with no conventional counterpart, in a literature that overwhelmingly assumes four-neighborhood movement and therefore never exercises diagonal paths. *Existing methods:* none surveyed target these directly; by definition they evade conventional test methods. Adjacent routing-speed work (e.g. [10], up to $\sim 72\%$ faster re-solving) only indirectly helps, by making re-routing around faults real-time tractable. *Drawbacks:* speed is not diagnosis — without a fault model for these classes, faster solving just repeats blind decisions.

2.2.6 Security-induced (cyberphysical) faults

Actuation tampering via the daisy chain, sensor spoofing, denial of service, sample and assay tampering, and IP piracy — treated in recent surveys as part of DMFB reliability alongside physical faults. *Existing methods:* surveyed as concerns (e.g. the 2025 MEDA review[11] and recent sustainability/security surveys[12] flag microelectrode degradation, excessive data-shift operations, and MEDA security exposure); dedicated countermeasures are outside the routing scope here. *Drawbacks for our purposes:* attacker-induced faults share the sensing surface (notably spoofing) but require a threat model this work does not claim.

2.2.7 Sensing-circuit faults

Each microcell’s actuate/sense circuitry is $0.35\,\mu\text{m}$ CMOS replicated ~ 1800 times per chip, so it inherits CMOS defect densities directly: the fault-recovery literature documents intra-MC defects reaching the sensing

circuits and builds backup readout multiplexers specifically so a faulty MC’s sensing result can be read through a neighbor. A stuck-at sense amplifier reporting “healthy” is phenomenologically our spoof class, and scan-chain read opens are our dropout class. Field exposure is worse: droplet sensing contributes up to 94% of all MC actuations [13], so the sense path wears first, while calibration drift, biofouling, and charge accumulation corrupt capacitive readings parametrically; the testing literature confirms sensing-circuit defects cause parametric *and* catastrophic failures that structural methods do not cover. Figure 2 shows the consequence: ground truth (left) versus the reported map (right) with spoofed cells (false positives), dropout cells (false negatives), and a misreported droplet position. *Existing methods*: component-level testing only (oscillation-based sensing tests, BIST built atop sensing circuits, sensitivity calibration) — no surveyed work models sensing faults as a first-class failure mode for routing; every sensing-map consumer treats the reported panel as ground truth. *Drawbacks of the status quo*: a fluidic fault masked by a simultaneous sensing fault on the same cell becomes an invisible trap, exactly where trust-the-map routing fails hardest. This layer is our work’s target.

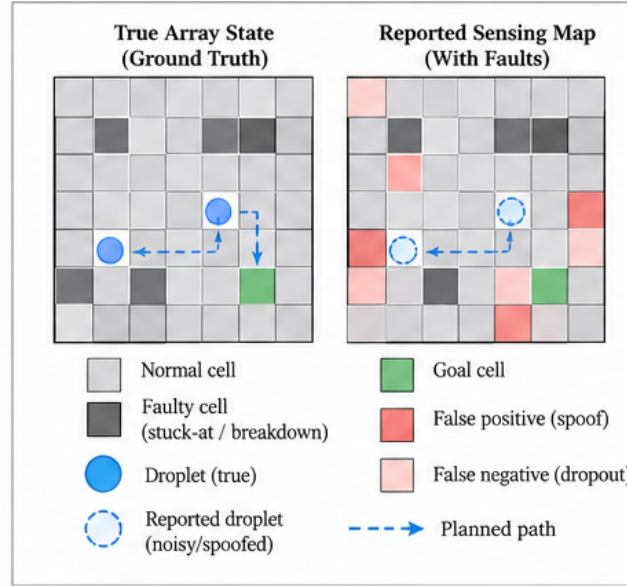


Figure 2: True array state (ground truth, left) versus the reported sensing map under sensing faults (right): spoofed healthy cells, dropout cells, and a misreported droplet position. Consumers of the sensing map see only the right panel.

The pattern across layers is consistent: each line of work owns one layer, and no surveyed method co-diagnoses control-path, sensing-circuit, and fluidic faults together.

3 Methodology

Figure 3 shows the closed loop in full implementation detail: a MEDA simulator (array dynamics, droplet kinematics, fault injection, sensing model, rewards) exchanges observations and rewards with a DQN agent whose inputs are the array state, the sensing map, the confidence map, and the task goal. All sweeps are configuration-only (`fault_profiles.yaml`); no experiment requires code changes.

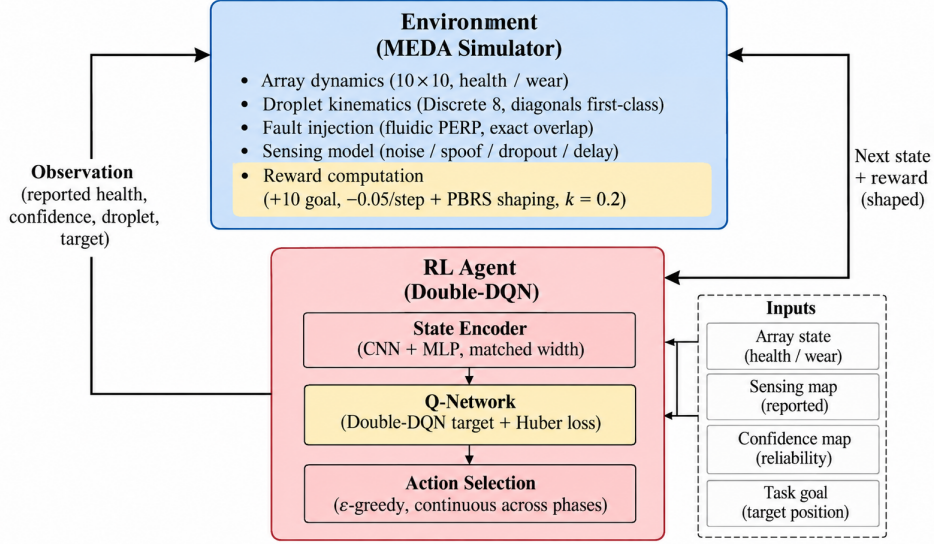


Figure 3: Agent-environment closed loop at implementation level: per-module simulator settings (array 10×10 , Discrete(8), decoupled faults, four sensing modes, shaped rewards) and the shared Double-DQN agent (matched encoder, Double target with Huber loss, continuous exploration); highlighted boxes are fix-cycle additions, identical across conditions except the confidence input.

3.1 Simulation environment

The array is a grid of microcells (default 10×10 , configurable), each holding ground-truth health $h \in [0, 1]$, a fault tag, cumulative wear, and droplet occupancy. The single droplet occupies one cell with a commanded target cell. Actuation offers eight directional moves (Discrete(8)): unlike the classic DMFB literature, which overwhelmingly assumes four-neighborhood (Manhattan) movement, MEDA’s free-direction routing makes diagonals first-class actions the agent selects among, and the simulator models them as such. Figure 4 shows the physical mechanism behind these moves. A move into a cell whose true health is at or below 0.3 silently fails (droplet stays put), modeling insufficient electrowetting force rather than a crash; out-of-bounds moves fail likewise. Diagonal moves validate only the destination cell, not the two orthogonal intermediaries (corner-cutting permitted): a deliberate modeling choice, flagged here because conventional DMFB routers typically forbid it, whose impact on learned policies is left for future ablations. Every actuated cell accrues wear (0.01/step), degrading health monotonically and progressing into dielectric breakdown at wear 1.0 — degradation and breakdown are one process, matching taxonomy (B). Array dimensions, voltage ranges ($\sim 20\text{--}40$ V), and droplet volumes are chosen within MEDA-literature ranges and recorded in `array_default.yaml`; the dynamics abstract actuation to the health-threshold model above.

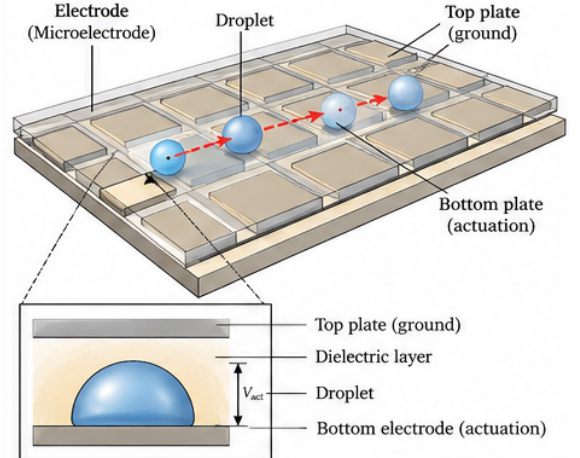


Figure 4: Electrowetting-on-dielectric actuation over a 2-D electrode array (cross-section inset): the physical mechanism behind the eight-directional moves the simulator models, including diagonals that classic four-neighborhood DMFB models omit.

3.2 Decoupled fault injection

Fluidic faults act *only* on ground-truth health: dielectric breakdown and stuck-at (catastrophic, permanent, $h=0$) and charge trapping (parametric, cumulative). Sensing faults act *only* on reported readings, never on ground truth: Gaussian noise ($\sigma=0.05\text{--}0.10$) on health; spoof reporting healthy (1.0) regardless of truth; dropout reading 0.0 (“no signal”) with zero-order hold on position; delay replaying lagged readings (position holds last reported — a documented freeze approximation). The four modes are catalogued in Figure 5. A per-episode scheduler samples the fluidic dead-cell set and the corrupted-sensor set *independently*, with the overlap fraction an explicit parameter (exact, not a lower bound): the regime the trust-the-map baseline handles worst. Fault density, type mix, and overlap live in `fault_profiles.yaml` (clean / low / medium / high / co-occurrence, plus a 0–50% density sweep); start and target cells are excluded from both sets.

3.3 Sensing-reliability (confidence) signal

The novel component. The droplet’s expected next position is derivable from the last commanded actuation; persistent deviation of the *reported* position implicates that cell’s sensor, independently of electrode health. Per cell we keep a rolling window (5) of residuals $r=\|\text{expected} - \text{reported}\|_2$ and report

$$c = \frac{1}{1 + \bar{r}} \in (0, 1], \quad (1)$$

so one-off noise washes out while persistent spoofing or freezing drives confidence down; all other cells relax toward 1 each step. The tracker update is skipped when the decision-time *reported* map already marks the destination dead (the residual then carries no position-sensor information) — a reported-only gate, never ground truth. The agent receives c as an input channel and *learns* the trust weighting; there is no hard-coded cutoff, keeping the baseline comparison fair. Figure 6 diagrams the full pipeline from capacitance computation to the confidence map.

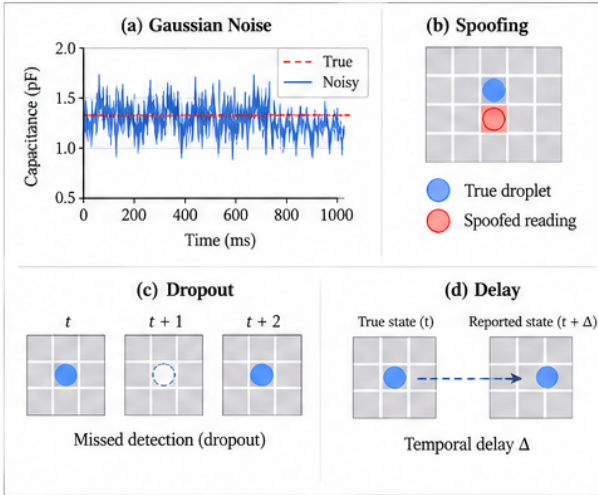


Figure 5: The four simulated sensing-fault modes: Gaussian health noise, spoofed (false-healthy) readings, dropout with missed detections, and temporal delay. Each corrupts only the reported plane, never ground truth.

3.4 Agents

Baseline and confidence-aware agents are the *same* Double-DQN implementation (online argmax, target evaluation [14], Huber loss, Adam, target sync every 500 gradient steps) with identical hyperparameters ($lr=10^{-3}$, $\gamma=0.99$, buffer 20000, batch 64, $\epsilon: 1.0 \rightarrow 0.05$ over 800 episodes) and matched depth/width: a two-layer CNN

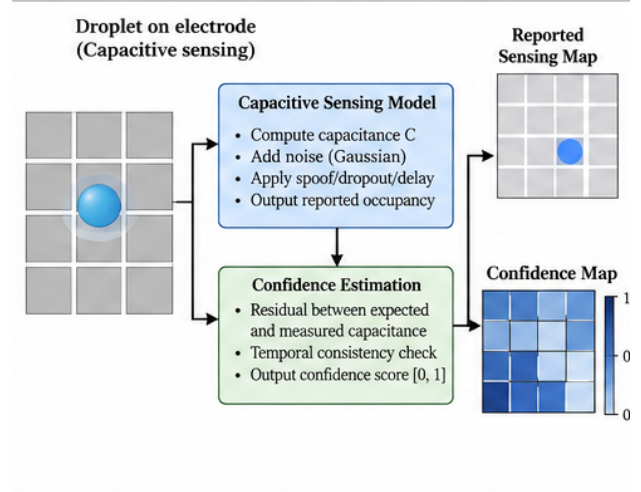


Figure 6: Simulated capacitive-sensing pipeline (left): capacitance computation with noise/spoof/dropout/delay injection yields the reported map, while the expected-vs-reported residual with temporal consistency yields the confidence map in $[0, 1]$ (right).

encoder with adaptive pooling plus MLP head. DQN is chosen deliberately: the prior deep RL routing lineage [1] is DRL-based, so the baseline must come from the same family for the comparison to mean anything; eight discrete moves suit value-based methods without continuous-control machinery; and single-agent scope defers multi-agent coordination overhead until the core sensing result is proven. The *only* model difference is input width (3 channels baseline: reported health, droplet, target; 4 aware: plus confidence), and the only environment difference is the confidence channel with its small suspect penalty. Rewards: +10 goal, −0.05/step, −1.0 for attempting a ground-truth-dead cell, −0.1 (aware only) for entering a sub-0.5-confidence cell. Dense guidance comes from potential-based shaping [15], $F = \gamma\Phi(s') - \Phi(s)$ with $\Phi = -k \cdot d_{\text{Chebyshev}}$, $k=0.2$ — policy-invariant, hence comparison-preserving — computed on *reported* positions, i.e. a function of the observation, consistent with the paper’s thesis.

3.5 Training curriculum and evaluation protocol

Phase A trains 1500 episodes under clean sensing (parity sanity check); phase B trains 1500 under a fixed medium sensing-fault rate so failures are encountered during learning. Phase 2 is *training*, not fine-tuning: network weights transfer across the boundary while the replay buffer is cleared, and one continuous epsilon schedule spans both phases. Both agents share paired curricula (identical reset-seed sequences) and paired greedy probes (every 250 episodes, $n=100$ for $\sim\pm 5\text{pp}$ selection noise); best-by-success checkpoints are saved and evaluation defaults to them, with fail-closed loading. Metrics are routing success rate, degradation versus sensing-fault density, false-trust rate (moves toward ground-truth-dead cells reported healthy), steps-to-goal, and greedy decision latency (warmed-up forward passes, reported with hardware as a ratio, not an absolute). The four-experiment battery (clean parity; 0–50% fault sweep; confidence ablation; co-occurrence) is defined in configuration; its numbers belong to the full report, not this interim one. Sweep densities are stress-test ranges, not claimed field rates.

4 Implementation and Results

4.1 Implementation map

The methodology of Section 3 is implemented as `meda-sensing-aware-rl/`: `meda_env/` (array, droplet, eight-directional (`Discrete(8)`) actuation with silent failure, decoupled `fault_injection/` and `sensing/`, Gymnasium wrapper), `agents/` (one shared Double-DQN class; baseline and aware differ only in input width), `training/` (two-phase curriculum with buffer clearing, continuous epsilon, paired seeds, greedy probes with best-checkpoint selection, fail-closed loading), `evaluation/` (success, degradation, false-trust, warmed-up decision latency), `experiments/` (parity, fault sweep, confidence ablation, co-occurrence), and `configs/` (all sweeps configuration-only). Thirty-six unit tests pin the load-bearing invariants: reported-position observations with no ground-truth leak, exact fault overlap, per-episode RNG seeding with pinned spawn order, scripted-action lockstep across agents, tracker gating semantics, and shaping-sign correctness. Figure 3 maps the same loop at implementation level: every fix-cycle addition is highlighted, and the two conditions share all unhighlighted boxes.

4.2 Setback: the first full-budget run failed to learn

The first 1500+1500-episode run, under the original sparse reward (+10 goal, −0.05/step) and vanilla DQN (MSE loss, max-bootstrap target), learned nothing: greedy probes peaked at 0.19 and decayed, with clean-sensing eval at 4.5% baseline / 11.5% aware — parity of failure. The loss curves split the diagnosis. The baseline diverged catastrophically (0.3→52 clean, spiking to 12441 under faults): rare +10 transitions arriving as massive TD errors inside a max-bootstrap feedback loop. The aware agent stayed numerically bounded (0.01–0.5) but its Q-values collapsed to the timeout value — a separate pathology, same outcome. The environment was exonerated first: a scripted Chebyshev-greedy policy scores 0.92 clean and 0.72 under medium sensing faults, so a learnable policy exists and the fault gap is real. The failure was learning configuration, and no fault-sweep numbers were drawn from these checkpoints: flat $\sim 5\%$ lines would have read as “no gap” when the truth was “no learning.”

4.3 Fixes

Three minimal, claim-preserving changes, applied identically to both agents:

- potential-based shaping ($k=0.2$, γ matched at 0.99) on *reported* positions, adding dense per-step gradient while preserving the optimal policy;
- Double-DQN target plus Huber loss, bounding the per-element gradient ($\delta=1$) and removing max-over-target feedback, with the learning rate untouched at 10^{-3} to isolate the effect;
- no architecture change, so any gain cannot be attributed to capacity.

Shaping signs and outlier-batch finiteness are unit-tested. Config-level diffs before/after (epsilon continuity, best-checkpoint selection, buffer clearing, $n=100$ probes) were verified identical across agents except for the observation channel — the earlier config audit found no confound.

4.4 Gated pilot evidence

A 300-episode clean pilot graded RED (final greedy probes 0.12/0.08), but against an $\varepsilon \approx 0.64$ schedule the probes measured a barely exploited policy, so the verdict was confounded rather than conclusive. Two sanity checks then passed: greedy evaluation is pure argmax with epsilon bypassed, and Q-spread over 20 states is healthy (median per-state Q-std ~ 0.25 , $\sim 7\times$ the random-init reference, argmax spread over 6–7 of 8 actions) — the encoder differentiates actions and TD targets flow, ruling out flat-Q collapse. The 900-episode extension (through full ε -decay) showed rising trajectories; the 1200-episode continuation, graded on trajectory, gave: aware 0.54 $\rightarrow$ 0.60 $\rightarrow$ 0.56 (GREEN: endpoint $\geq 40\%$, rise-then-plateau convergence) and baseline 0.46 $\rightarrow$ 0.42 $\rightarrow$ 0.36 (amber churn around ~ 0.40 with converged loss). Figure 7 records the full 12-probe trajectories against the 15/25/40% gate bands and the 0.92 scripted ceiling; Figure 8 the monotonically converging Huber losses (max 0.15 versus 12441 pre-fix); Figure 9 the behavior/exploration split explaining the early red; Figure 10 the Q-spread health check.

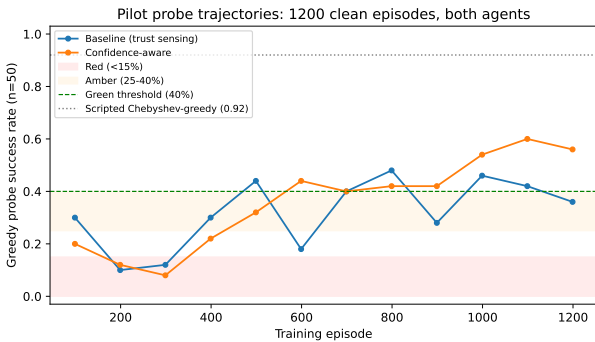


Figure 7: Greedy probe trajectories over 1200 clean episodes ($n=50$), both agents, with gate bands and the scripted-greedy ceiling.

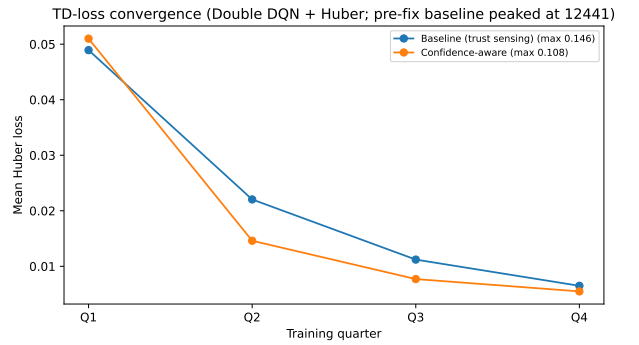


Figure 8: TD-loss quarter means, both agents. Bounded, monotonically converging losses are the stability proof that separates this run from the failed one (pre-fix baseline peaked at 12441); without them, rising probes could be noise rather than learning.

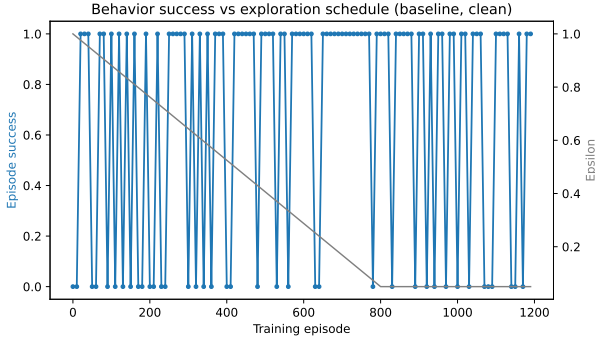


Figure 9: Train-time (exploratory) success versus the epsilon schedule (baseline). Behavior succeeds on exploration luck while greedy probes lag, which is why premature probes mislead — this split is what made the 300-episode RED verdict confounded rather than conclusive.

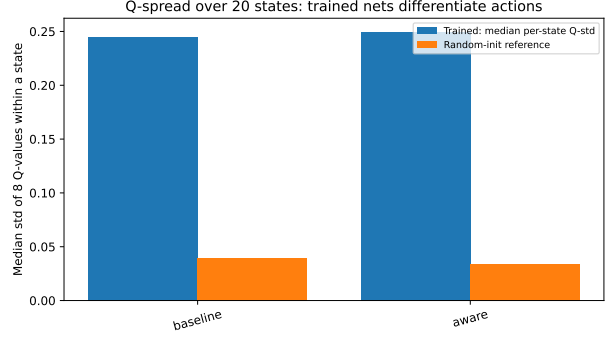


Figure 10: Median per-state Q-value std over 20 states versus random-init reference. If Q were flat (all actions valued alike), the encoder would be the suspect and the fallback would change; the $\sim 7\times$ spread rules that out and pins the earlier failure on TD targets instead.

4.5 Limits of the current evidence

Three limits bound what is claimed. First, all runs above are clean-sensing phase-A equivalents: faulty-phase training and experiments 1–4 belong to the full retrain, and no fault-robustness gap is claimed here. Second, the baseline’s late probe churn ($\sim \pm 8\text{pp}$ around 0.40 at $n=50$) is unresolved detail — converged loss with a fluctuating greedy policy — which the full run’s $n=100$ probes and best-checkpoint selection are designed to settle. Third, pilot probes are single-seed (paired across agents, but one seed); multi-seed confidence intervals are deferred to the full evaluation. Fourth, all training and evaluation use the 10×10 default grid: whether the confidence signal survives at commercial array sizes, where prior deep RL routers operate at commercial scale [1], is untested and reserved for follow-up scaling studies. Full training (1500 clean plus 1500 faulty episodes, $n=100$ probes, best-checkpoint selection) is expected to raise clean-parity levels toward the scripted ceiling and to produce the faulty-phase gap that is this project’s headline; no numeric values are asserted before that run. A final exclusion, stated for the record: a composite project figure containing anticipated (not measured) fault-sweep bars was deliberately left out of this report — every number above traces to logged data.

5 Conclusion and Future Works

Prior RL routing for MEDA biochips is fault-aware for the fluidic layer and blind to the sensing layer: it treats a possibly corrupted sensing map as ground truth. This project relaxes that assumption with a router that receives a per-cell residual-based confidence signal and learns how much to trust each reading, evaluated in a fully digital simulator whose fluidic and sensing fault channels are architecturally decoupled. The matched baseline/aware design isolates any performance gap to learned distrust rather than capacity.

This interim report established the precondition for that comparison. The first full-budget run failed to learn under sparse rewards and vanilla DQN dynamics (baseline loss peaking at 12441); minimal, identically applied fixes — potential-based shaping on reported positions, Double-DQN target, Huber loss — restored stable learning with monotonically converging losses; and gated pilots gave GREEN on clean sensing (confidence-aware plateau $0.42 \rightarrow 0.56$; baseline churning around 0.40 with converged loss), clearing the way for the full retrain. No fault-robustness gap is claimed here: faulty-phase training and experiments 1–4 are the next milestone, with checkpoints, TensorBoard logs, and fail-closed evaluation already wired for it. The immediate next step is completing the full 1500+1500-episode retrain for both agents and running experiments 1–4 against it; beyond that, we plan systematic optimization — reward-scale and shaping-coefficient studies, hyperparameter sweeps around the stabilized Double-DQN, and the MLP ablation if encoder capacity ever binds — before any architectural change.

Beyond the headline comparison, the queued work is: multi-seed evaluation with confidence intervals (pilot probes are paired but single-seed); the flat-MLP-over-one-hots ablation if the CNN’s smeared single-

pixel features ever limit performance; a voltage-to-threshold actuation mapping to replace the current health-threshold abstraction; trailing (rather than frozen) position-delay sensing; grounding sweep densities to CMOS defect and calibration-drift rates instead of stress-test ranges; and, longest term, multi-droplet extension and physical validation of the sim-to-real gap — specifically, calibrating the abstracted health-scalar sensing model against measured capacitive readout noise, drift, and fault rates from fabricated hardware, without which the transfer risk of learned trust policies remains open. Each of these — scale-up, corner-cutting ablations, and hardware-grounded sensing calibration — is declared here as actively planned exploration, not background possibility.

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